Ecological Niche Modeling of *Vespa velutina*. Methodology, Theory, and Research Assets Part I

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1 Introduction

This is not part of my work anywhere, I just do it for fun

2 Vespa velutina

3 Model

4 Getting presences

5 Generating pseudo-absences

To effectively model the potential niche of the *Vespa velutina*, the algorithm must distinguish between suitable and unsuitable habitats. This requires both presence and absence data. However, obtaining true absences (ground truth) is logistically challenging and expensive, because not sighting a hornet (or any other species) does not confirm its absence. Therefore, the standard practice is to generate pseudo-absences: synthetic points where the species is assumed to be absent.

While presence-only models like Maxent are widely utilized—often relying on "background data" that can lead to misleading interpretations of a species' actual niche—current literature consistently demonstrates that presence-absence frameworks yield superior predictive performance. However, the generation of pseudo-absences is a non-trivial process. Stochastic or random sampling often fails to encapsulate the complex spatial and environmental dynamics of an active invasion, even though they can perform better in more stable species.

To address this, the present study adopts a dual-framework approach. We baseline our methodology on the seminal principles established by [**?**] in *'Selecting pseudo-absences for species distribution models: how, where and how many?'*, while simultaneously integrating the advanced Three-Step Selection (SM4) technique proposed by [**?**].

By synthesizing the statistical robustness of the former with the environmental and geographic filtering of the latter, this project evaluates multiple pseudo-absence configurations. This comparative framework allows for a high-resolution exploration of how different sampling strategies influence model sensitivity, specificity, and ultimately, the reliability of the invasive risk maps for *Vespa velutina*.

5.1 Beyond Static Modeling: Implementing the SM4 Protocol

While our initial approach using the Barbet-Massin method provided a robust baseline, modeling the distribution of Vespa velutina requires a more surgical selection of pseudo-absences. Although the hornet has been present in Western Europe for over two decades, its expansion remains a dynamic process. The sharp frontier currently observed (particularly between the Atlantic regions and the Castilian Plateau) might suggest a finished invasion; however, it is more likely a reflection of steep environmental gradients where the species faces immediate physiological barriers, such as extreme hydric stress and thermal limits.

The implementation of the SM4 (Four-Step Metadata-derived Multi-strategy) protocol is essential to address this nuance. Unlike broader methods, SM4 optimizes the spatial and environmental distance of pseudo-absences, ensuring they are not selected from ecologically obvious areas (which artificially inflates accuracy) nor from

the species' immediate expansion front. This prevents the model from mistaking a temporary dispersal bottleneck for true habitat unsuitability. By shifting to this multi-stage strategy, we expect to significantly reduce spatial autocorrelation and force the Random Forest to discriminate between truly habitable zones and those at the edge of the hornet's physiological threshold. This refinement allows us to determine if the current borders are absolute biological walls or merely 'porous' corridors.

However, to fully appreciate the necessity of this integrated approach, it is essential to first evaluate the individual components that define it. Therefore, this study explicitly tests the SM1, SM2, and SM3 protocols as a series of incremental benchmarks. This systematic evaluation is not merely a comparative exercise, but a fundamental requirement to isolate and quantify the specific contribution of spatial and environmental constraints on model accuracy. By analyzing these intermediate stages, we can decouple the effects of geographic proximity from environmental filtering, thereby establishing a diagnostic baseline. This stepwise approach allows for a rigorous validation of the SM4 framework, demonstrating that its superior predictive power is the result of an integrated synergy rather than an artifact of any single filtering process. Consequently, this comparative hierarchy serves to justify the transition from stochastic sampling to an ecologically informed selection of pseudo-absences, ensuring that the final risk maps for Vespa velutina are both statistically robust and biologically plausible.

5.1.1 A simple random selection: SM1

The Simple Random Pseudo-absence Selection (SM1) serves as the baseline approach in this study. In this method, pseudo-absence points are generated by randomly sampling locations across the entire study area. A critical 1-km spatial exclusion constraint was applied to ensure that no pseudo-absence is selected within the same grid cell as a known presence record. This alignment with the environmental predictor resolution (1 km) is fundamental, because it prevents the model from receiving conflicting signals for the same pixel, which would otherwise degrade the algorithm's learning process. Furthermore, this thinning ensures spatial independence by avoiding redundant data within the same 1-km pixel. While computationally straightforward, SM1 assumes that the species' absence is equally likely across all unsampled locations. However, as noted in the literature, this method is highly susceptible to sampling bias and may inadvertently include suitable but unoccupied habitats (false absences), potentially leading to a degradation in the model's ability to discriminate between presence and absence.

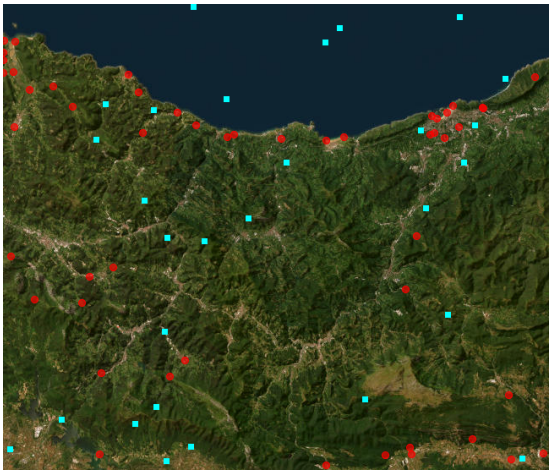

Figure 1: *Local distribution of Vespa velutina presences (red) and SM1 pseudo-absences (blue) in Gipuzkoa, Spain.*

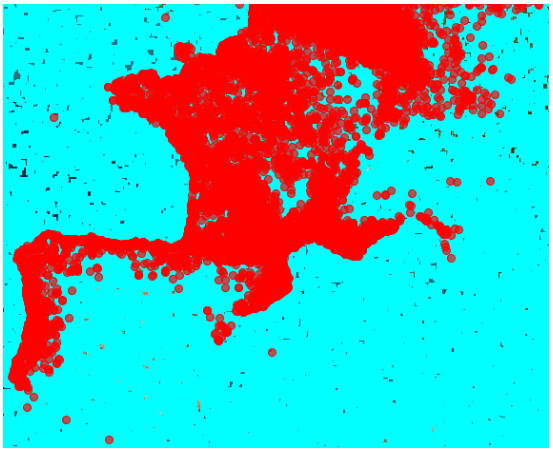

Figure 2: *Regional extent of presences (red) and SM1 pseudo-absences (blue) across Western Europe.*

5.1.2 Constraining space: SM2

6 Generating pseudoabsences

7 Metrics